



Week 03 Task-09

Enumeration Digging Deeper into Services

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1) Executive Summary

This task focused on performing service enumeration to gather valuable information from accessible network services. FTP and SSH banner grabbing were conducted to identify service names and version details. SMB enumeration was performed using enum4linux to discover available shares, users, and other relevant SMB information. The techniques were also practiced in the TryHackMe Vulniversity room to strengthen practical enumeration skills and privilege escalation. Other than that Nmap tryHackme room was also completed to practice nmap scanning techniques. The findings demonstrate the importance of proper service enumeration as an essential phase of penetration testing.

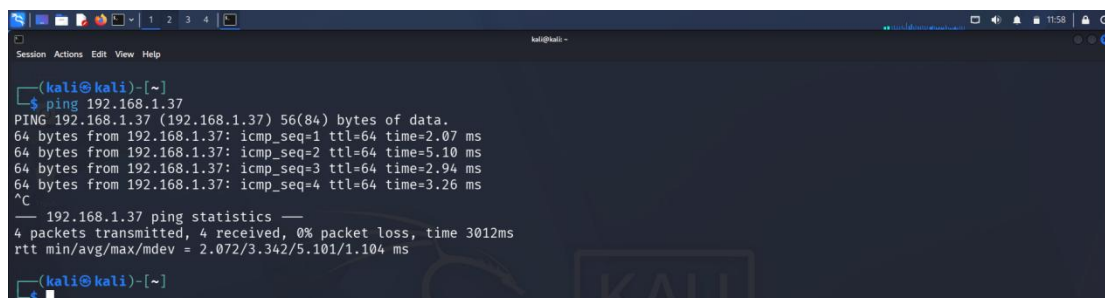
2) Scope of Task

The scope of this task was limited to FTP, SSH, and SMB service enumeration in an authorized lab environment on home-based network. Banner grabbing and enum4linux were used to collect service versions, banners, shares, and user-related information. The task also included practical hands-on learning through the TryHackMe Vulniversity room and Nmap room.

3) Tools Used

- Kali Linux
- Nmap
- enum4linux
- netcat

4) Verifying Target Connectivity

A screenshot of a Kali Linux terminal window. The prompt is (kali@kali)-[~]. The user has entered the command \$ ping 192.168.1.37. The output shows four successful ping responses from 192.168.1.37 with varying times (2.07 ms, 5.10 ms, 2.94 ms, 3.26 ms). Below the responses, a summary line shows '192.168.1.37 ping statistics' and '4 packets transmitted, 4 received, 0% packet loss, time 3012ms'. The terminal window has a dark background and a title bar at the top.

In the above figure **ping 192.168.1.37** is executed and it is confirmed that target Host is reachable the results are explained below

Observed Results

Target IP:	192.168.1.37
Packets Sent:	4
Packets received:	4
Average Response Time:	3.34ms
Host Status:	reachable

5) SMB Enumeration

Observed Results

```
Session Actions Edit View Help
kali@kali: ~
$ enum4linux -a 192.168.1.34
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Sat Sep  5 22:16:40 2026

( Target Information )
Target ..... 192.168.1.34
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

( Enumerating Workgroup/Domain on 192.168.1.34 )
[+] Got domain/workgroup name: WORKGROUP

( Nbtstat Information for 192.168.1.34 )
Looking up status of 192.168.1.34
METASPLOITABLE <00> - B <ACTIVE> Workstation Service
METASPLOITABLE <03> - B <ACTIVE> Messenger Service
METASPLOITABLE <20> - B <ACTIVE> File Server Service
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections
MAC Address = 00-00-00-00-00-00

( Session Check on 192.168.1.34 )
[+] Server 192.168.1.34 allows sessions using username '', password ''
```

The **enum4linux -a** command was executed against the target system to enumerate SMB information, including system details and available network resources.

```
Session Actions Edit View Help
kali@kali: ~
( Share Enumeration on 192.168.1.34 )

Sharename      Type      Comment
-----
print$         Disk      Printer Drivers
tmp            Disk      oh noes!
opt            Disk
IPC$           IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
ADMIN$         IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))

Reconnecting with SMB1 for workgroup listing.

Server          Comment
-----
Workgroup       Master
WORKGROUP

[+] Attempting to map shares on 192.168.1.34
//192.168.1.34/print$ Mapping: DENIED Listing: N/A Writing: N/A
//192.168.1.34/tmp Mapping: OK Listing: OK Writing: N/A
//192.168.1.34/opt Mapping: DENIED Listing: N/A Writing: N/A

[E] Can't understand response:
NT_STATUS_NETWORK_ACCESS_DENIED listing \*
//192.168.1.34/IPC$ Mapping: N/A Listing: N/A Writing: N/A
//192.168.1.34/ADMIN$ Mapping: DENIED Listing: N/A Writing: N/A

( Password Policy Information for 192.168.1.34 )
Password:
```

The enumeration results revealed the target computer name, workgroup, Samba version, user accounts, and available SMB shared folders.

```

kali@kali ~
Session Actions Edit View Help
[+] Attaching to 192.168.1.34 using a NULL share
[+] Trying protocol 139/SMB ...
[+] Found domain(s):
    [+] METASPLOITABLE
    [+] Builtin
[+] Password Info for Domain: METASPLOITABLE
    [+] Minimum password length: 5
    [+] Password history length: None
    [+] Maximum password age: Not Set
    [+] Password Complexity Flags: 000000
    [+] Domain Refuse Password Change: 0
    [+] Domain Password Store Cleartext: 0
    [+] Domain Password Lockout Admins: 0
    [+] Domain Password No Clear Changes: 0
    [+] Domain Password No Anon Change: 0
    [+] Domain Password Complex: 0
    [+] Minimum password age: None
    [+] Reset Account Lockout Counter: 30 minutes
    [+] Locked Account Duration: 30 minutes
    [+] Account Lockout Threshold: None
    [+] Forced Log off Time: Not Set
[+] Retrieved partial password policy with rpcclient:

```

```

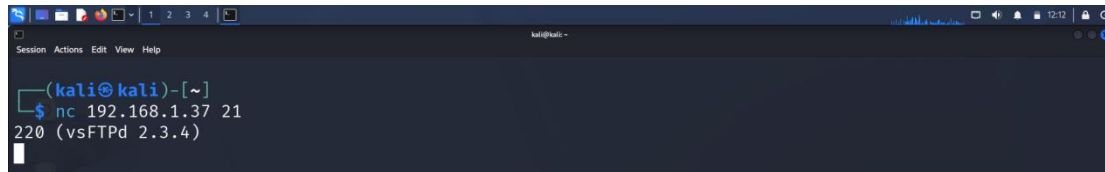
kali@kali ~
Session Actions Edit View Help
( Groups on 192.168.1.34 )
[+] Getting builtin groups:
[+] Getting builtin group memberships:
[+] Getting local groups:
[+] Getting local group memberships:
[+] Getting domain groups:
[+] Getting domain group memberships:
( Users on 192.168.1.34 via RID cycling (RIDS: 500-550,1000-1050) )
[!] Found new SID:
S-1-5-21-1042354039-2475377354-766472396
[+] Enumerating users using SID S-1-5-21-1042354039-2475377354-766472396 and logon username '', password ''
S-1-5-21-1042354039-2475377354-766472396-500 METASPLOITABLE\Administrator (Local User)
S-1-5-21-1042354039-2475377354-766472396-501 METASPLOITABLE\nobody (Local User)
S-1-5-21-1042354039-2475377354-766472396-512 METASPLOITABLE\Domain Admins (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-513 METASPLOITABLE\Domain Users (Domain Group)

```

Observed Results

Parameter	Result
Target IP	192.168.1.34
Computer Name	Metasploitable
Workgroup	WORKGROUP
SMB Version	Samba 3.0.20-Debian
User accounts	root, msfadmin, mysql, ftp, www-data and others
Shared Folders	Print\$, tmp, opt, IPC\$, ADMIN\$
Accessible share	tmp

6) FTP Banner Grabbing



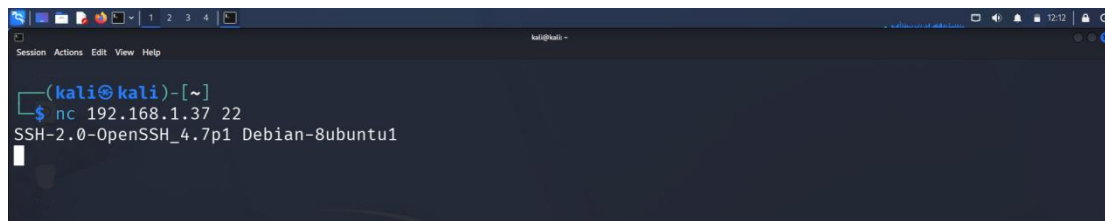
```
(kali㉿kali)-[~]  
$ nc 192.168.1.37 21  
220 (vsFTPd 2.3.4)
```

Banner Grabbing is the process of retrieving information such as a service name, software version or operating system on a target network which will help penetration testers to identify vulnerabilities and better understand the attack surface. In the above Figure FTP Banner Grabbing is performed using netcat tool . command used is **nc 192.168.1.37 21**. Information obtained from banner is explained below.

Observed Results

Target IP address:	192.168.1.37
Port Number:	21
Service:	FTP
Banner:	220(vs FTPd 2.3.4)
FTP Software:	vsFTPd 2.3.4

7) SSH Banner Grabbing



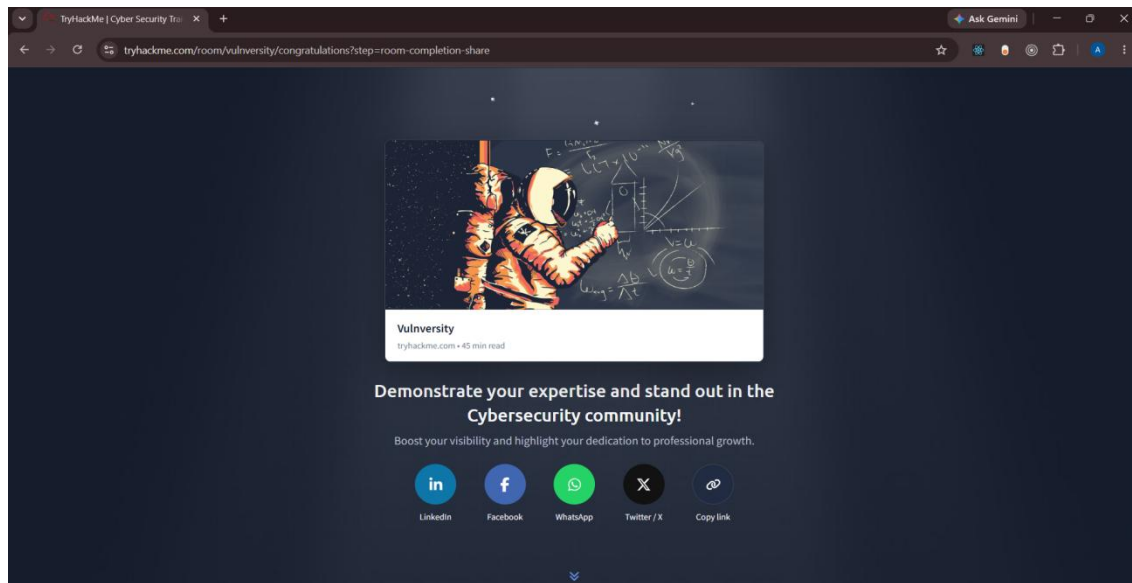
```
(kali㉿kali)-[~]  
$ nc 192.168.1.37 22  
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
```

Banner Grabbing is the process of retrieving information such as a service name, software version or operating system on a target network which will help penetration testers to identify vulnerabilities and better understand the attack surface. In the above Figure SSH Banner Grabbing is performed using netcat tool. command used is **nc 192.168.1.37 22**. Information obtained from banner is explained below.

Observed Results

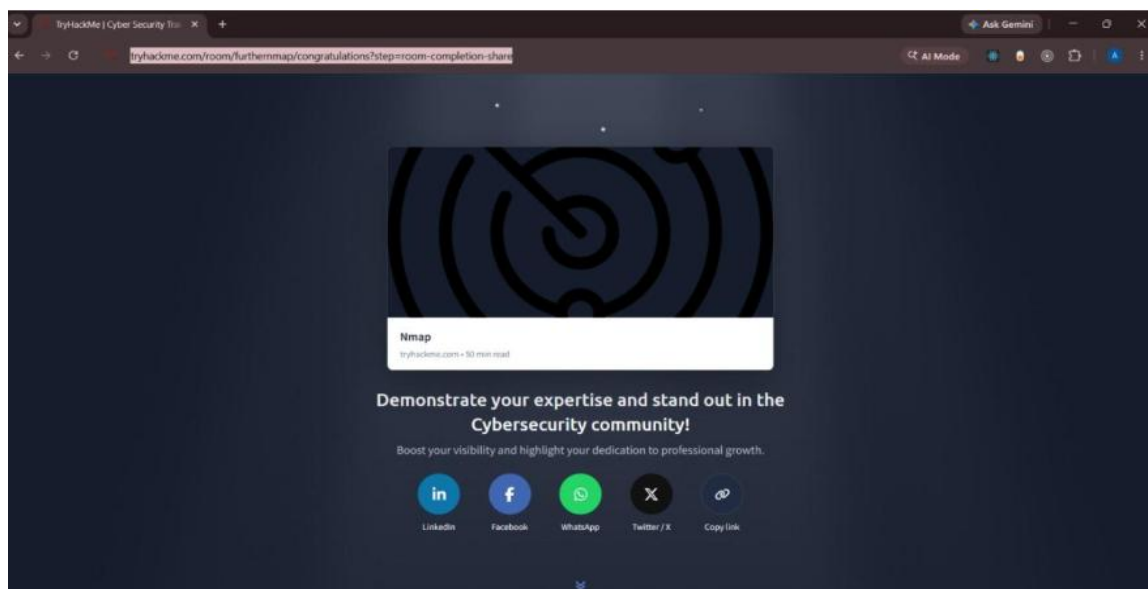
Target IP address:	192.168.1.37
Port Number:	22
Service:	SSH
Banner:	SSH-2.0- OpenSSH_4.7p1 Debian-8ubuntu1
SSH Software:	OpenSSH 4.7p1 Debian-8ubuntu1

8) TryHackme room vulniversity



As part of the practical learning, the TryHackMe Vulniversity room was completed. The exercise involved performing reconnaissance and enumeration, discovering hidden directories using Gobuster, and identifying a path to compromise the web server. After gaining initial access, privilege escalation techniques were performed to obtain higher-level access, demonstrating the complete penetration testing process from reconnaissance to system compromise.

9) TryHackme room Nmap



TryHackMe room Nmap was completed. This room provides me hands-on experience with different network or target scanning techniques such as host discovery, port scanning, enumeration, stealth scan NSE scripts for vulnerability scanning, tcp scan, xmass scan and so on.

10) Conclusion

This task provided me practical experience in service enumeration, banner grabbing, and SMB enumeration using enum4linux. The Vulniversity room further strengthened my skills in reconnaissance, web server exploitation, and privilege escalation. I also performed Nmap practice on TryHackme room Overall, the task demonstrated the importance of proper enumeration in identifying potential attack paths during penetration testing.